26 MOTION IN THE UNIVERSE

We live on a planet called Earth. It is one of many planets that, together with their moons, form our Solar System. Gravitational forces hold our Solar System together. These forces cause the planets, asteroids and comets to orbit the Sun, and moons and artificial satellites to orbit the planets. In this chapter we will look at the key role played by gravitational forces in controlling these movements.

UNITS

In this section you will need to use kilogram (kg) as the unit of mass, metre (m) as the unit of length, metre/second (m/s) as the unit of speed, metre/second^2 (m/s^2) as the unit of acceleration, newton (N) as the unit of force, second (s) as the unit of time and newton/kilogram (N/kg) as the unit of gravitational field strength.



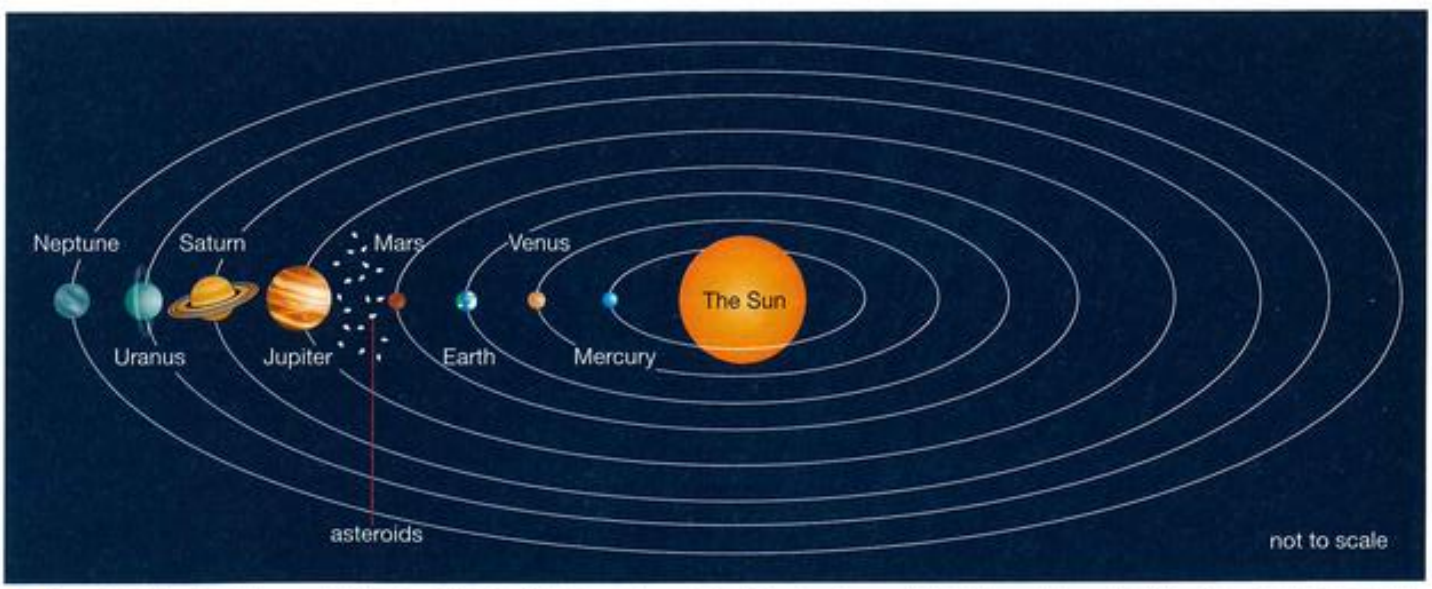
▲ Figure 26.1 The force of gravity keeps these moons in orbit around Jupiter.

LEARNING OBJECTIVES

- Explain that gravitational force causes moons to orbit planets; causes the planets and comets to orbit the Sun; causes artificial satellites to orbit the Earth
- Understand why gravitational field strength, g, varies and know that it is different on other planets and the Moon from that on the Earth
- Describe the differences in the orbits of comets, moons and planets
- Use the relationship between orbital speed, orbital radius and time period:

$$v = 2 \times \pi \times \frac{\text{orbital radius, } r}{\text{time period, } T}$$

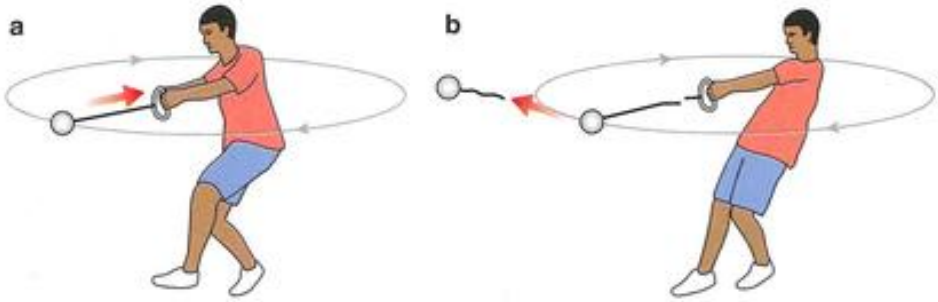
THE SOLAR SYSTEM



▲ Figure 26.2 The Earth is one of eight planets that orbit the Sun. The orbits of the planets are elliptical (an oval shape) with the Sun close to the centre.

WHY DO OBJECTS MOVE IN A CIRCLE?

Figure 26.3 shows a boy swinging a heavy ball around on a wire. To make this ball travel in a circle he needs to spin around and at the same time pull on the wire. Without this continuous pulling force the ball will not travel in a circle.



▲ Figure 26.3 **a** A 'pulling' force has to be applied to the ball to make it travel in a circle. **b** If the wire breaks or the boy releases the handle the ball flies away.

Planets and comets travel around the Sun. Moons and satellites travel around the planets. For this to happen there must be forces being applied to them. There is no string or wire to pull on as in the example above, so where do these forces come from? In 1687, Isaac Newton suggested his theory of gravity to explain these movements.

Newton suggested that between any two objects there is always a force of attraction. This attraction is due to the masses of the objects. He called this force gravitational force.

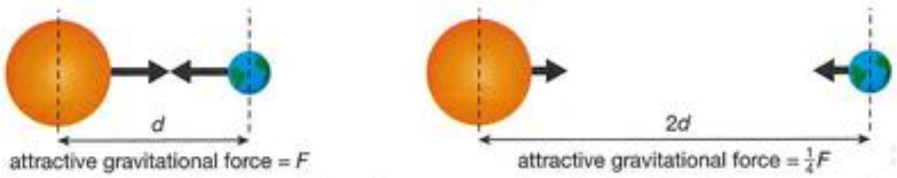
He suggested that the size of this force depends on the:

1 masses of the two objects
2 distance between the masses.



The greater the masses of the two objects the stronger the attractive forces between them.

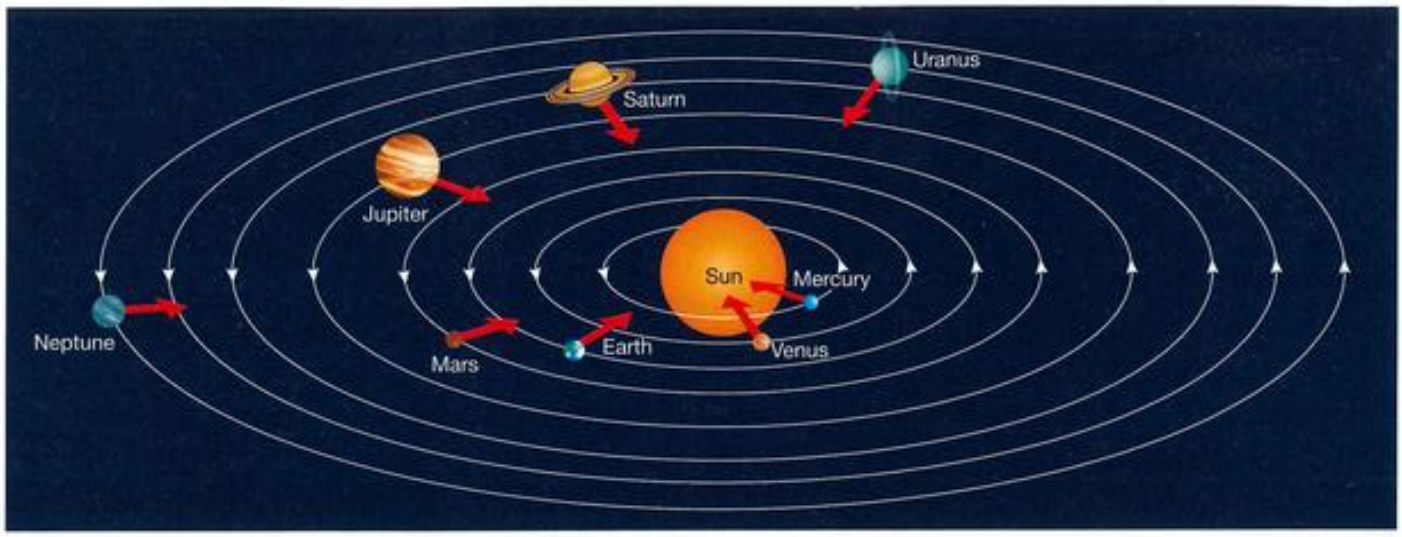
If the distance between the masses is increased the forces between them decrease.



▲ Figure 26.4 Gravitational forces obey an inverse square law – that is, if the distance between the masses is doubled, the forces between them are **quartered**; if the distance between them is **trebled**, the forces become one ninth of what they were.

The gravitational attraction between two objects with small masses is tiny. Only when one or both of the objects has a very large mass – for example, a moon or a planet – is the force of attraction obvious.

Our Sun is massive. It contains over 99% of the mass of the Solar System. It is the gravitational attraction between this mass and each of the planets that holds the Solar System together and causes the planets to follow their curved paths.



▲ Figure 26.5 Gravitational forces make the planets follow nearly circular paths.

Those planets that are closest to the Sun feel the greatest attraction and so follow the most curved paths. Planets that are the furthest from the Sun feel the weakest pull and follow the least curved path.

EXTENSION WORK

The planet Neptune is a very long way from the Sun so the gravitational force between them is very small. This means that the orbit of Neptune is not very curved and it takes a very long time for it to complete one orbit.

Planet	Average distance from Sun compared with the Earth	Time for one orbit of the Sun in Earth years
Mercury	0.4	0.2
Venus	0.7	0.6
Earth	1.0	1.0
Mars	1.5	1.9
Jupiter	5.0	12
Saturn	9.5	30
Uranus	19	84
Neptune	30	165



▲ Figure 26.6 Some of the many moons of Saturn.

SATELLITES AND MOONS

A satellite is an object that orbits a planet. There are two types of satellite: natural and artificial (human-made).

Natural satellites are called moons. The Earth has just one moon. It is the fifth largest moon in our Solar System, approximately 340 000 km from Earth, and takes just over 27 days to complete one orbit. Although we call our moon 'The Moon' it is not unique. Many planets have moons. Some have more than one, for example, Mars has two moons while Jupiter and Saturn have more than 60 each. All moons have circular orbits because of the gravitational forces between them and their planet.

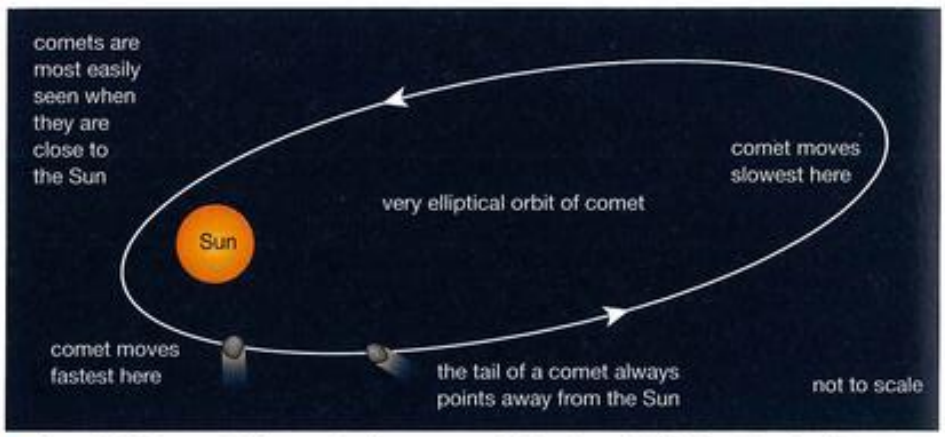


▲ Figure 26.7 This satellite is held in orbit by the gravitational attraction between it and the Earth.

Since the late 1950s humans have been able to launch, and to put into orbit around the Earth, objects like the one seen in Figure 26.7. These are known as artificial satellites and are extremely useful. Some satellites are put into a very high orbit above the Earth and are used to help us communicate over large distances, for example, for international phone calls or video links, the internet and so on. Some satellites are put into a much lower orbit and are used to monitor in detail the Earth's surface, such as the temperature of the world's oceans or the progress of forest fires.

COMETS

Comets are large rock-like pieces of ice that orbit the Sun. They have very elliptical (elongated) orbits which at times take them very close to the Sun. At other times they travel close to the very edge of our Solar System.



▲ Figure 26.8 The speed of a comet varies enormously. They travel at their fastest when they are very close to the Sun.

EXTENSION WORK

Although you will not be asked in your exam about the 'tails of comets' it is interesting to know that, close to the Sun, some of a comet's frozen gases evaporate to become gases again, forming a long tail that shines in the sunlight. These tails can be millions of kilometres in length. Perhaps the most famous of the comets is Halley's Comet, which visits our part of the Solar System every 76 years. It was last visible from Earth in 1986.



▲ Figure 26.9 Halley's comet could be last seen from Earth in 1986.

GRAVITATIONAL FIELD STRENGTH

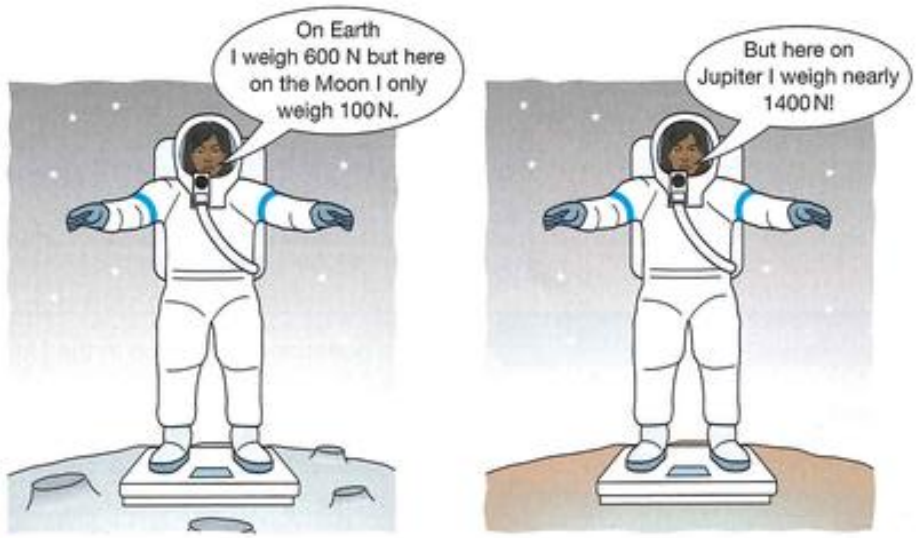
The strength of gravity on a planet or moon is called its gravitational field strength, and given the symbol g. Different planets have different masses and different radii – both of these will affect their gravitational field strengths.

- The larger the mass of a planet the greater its gravitational field strength.
- The larger the radius of a planet the smaller the gravitational field strength at its surface.

The gravitational field strength on the Earth is approximately 10 N/kg whilst on the Moon it is approximately 1.6 N/kg.

Planet	Diameter compared with the Earth	Mass compared with the Earth	Gravitational field strength/N/kg
Mercury	0.4	0.06	4
Venus	0.9	0.38	9
Earth	1.0	1.0	10
Mars	0.5	0.10	4
Jupiter	11	320	23
Saturn	9	95	9
Uranus	4	15	9
Neptune	4	17.0	11

On the Moon, the gravitational field strength is only one sixth that of the Earth's. This means that on the Moon you would weigh six times less. If however you visited Jupiter you would weigh almost 2½ times more!



▲ Figure 26.10 The difference in one person's weight on the Moon and on Jupiter

ORBITAL SPEEDS OF PLANETS AND SATELLITES

The **orbital speeds** of satellites vary greatly depending on the tasks they are performing. For example, communication satellites are put in high orbits and travel at approximately 3 km/s, while those observing the whole surface of the Earth are put into low polar orbits with speeds of about 8 km/s.

We can calculate the speed of a satellite using the equation:

$$\text{speed, } v = \frac{\text{distance moved, } s}{\text{time taken, } t}$$

The distance moved is the circumference of the circular orbit, r:

distance moved = $2 \times \pi \times$ orbital radius, r

The time period, T, is the time for one complete orbit.

EXTENSION WORK

The orbital period of a communications satellite, and most weather satellites, is 24 hours, and it always remains above the same part of the Earth's surface. It is said to be 'geostationary'.

REMINDER

Remember to use the same units throughout your work. Distances here are likely to be given in kilometres and so must be converted into metres. In this example, you must multiply both 8.2 km and 6600 km by 1000. Times given in hours and minutes need to be converted into seconds.

So:

$$\text{orbital speed, } v = \frac{2 \times \pi \times \text{orbital radius, } r}{\text{time period, } T}$$

$$v = \frac{2 \times \pi \times r}{T}$$

The same equation can be used to calculate the speeds of the planets, where r is the average distance from the Sun.

EXAMPLE 1

Calculate the speed of a satellite that is orbiting 200 km above the Earth's surface and completes one orbit in 1 h 24 min. The radius of the Earth is 6400 km.

The radius of the orbit is measured from the centre of the Earth – that is, it is the radius of the Earth plus the height of the orbit above the Earth's surface.

radius of orbit, r = 200 km + 6400 km
= 6600 km
= 6 600 000 m

orbital period, T = (60 min + 24 min) × 60 seconds
= 5040 seconds

$$\text{orbital speed, } v = \frac{2 \times \pi \times r}{T}$$

$$= \frac{2 \times \pi \times 6\,600\,000 \text{ m}}{5040 \text{ s}}$$

$$= \frac{41\,500\,000 \text{ m}}{5040 \text{ s}}$$

= 8200 m/s or 8.2 km/s

CHAPTER QUESTIONS

More questions on motion in the Universe can be found at the end of Unit 8 on page 275.

SKILLS CRITICAL THINKING (3, 4, 6)

1 a What is the name of the force that keeps all the planets of the Solar System in orbit around the Sun?

b What two factors determine the size of this force on the surface of a planet.

c Describe one difference between the orbits of Mercury and Neptune. Give one reason for this difference.

d Describe how the speed of a comet changes as it orbits the Sun.

2 Describe the differences between the orbits of a moon, a planet and a comet.

SKILLS PROBLEM SOLVING (10)

HINT

Convert the time period into hours.

3 A satellite is in an orbit 35 786 km above the surface of the Earth. The radius of the Earth is approximately 6400 km. The satellite is moving at 3.07 km/s.

a How long does the satellite take to orbit the Earth once?

SKILLS REASONING (7)

b What do you notice about the time period of this satellite?

SKILLS PROBLEM SOLVING (9)

4 If the Earth is approximately 150 million km from the Sun calculate the speed at which the following planets are moving around the Sun. The information you need is given in the table on page 261.

a Earth b Jupiter